

Reference

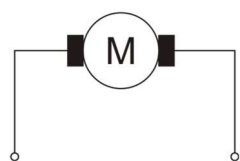
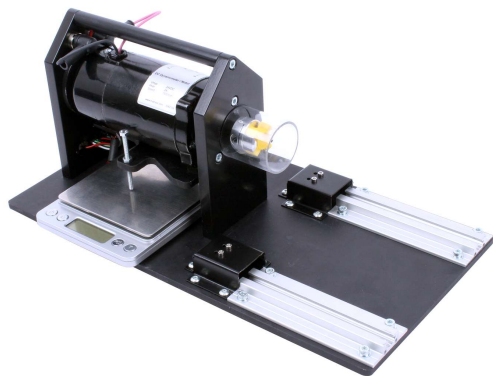
**Modern electrical
machines**

Machine information

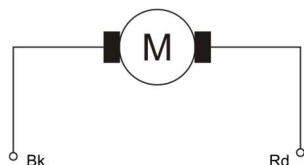
Reference

The dynamometer

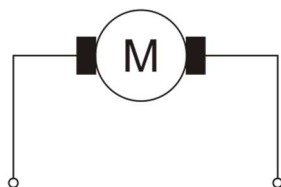
Modern electrical machines



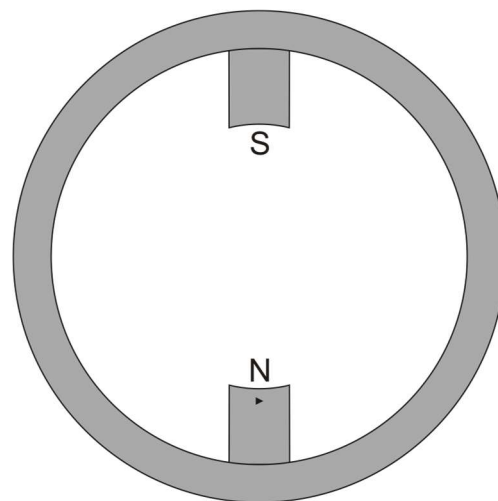
Conventional symbol



Actual circuit



Circuit in use



Stator winding circuit

Connection

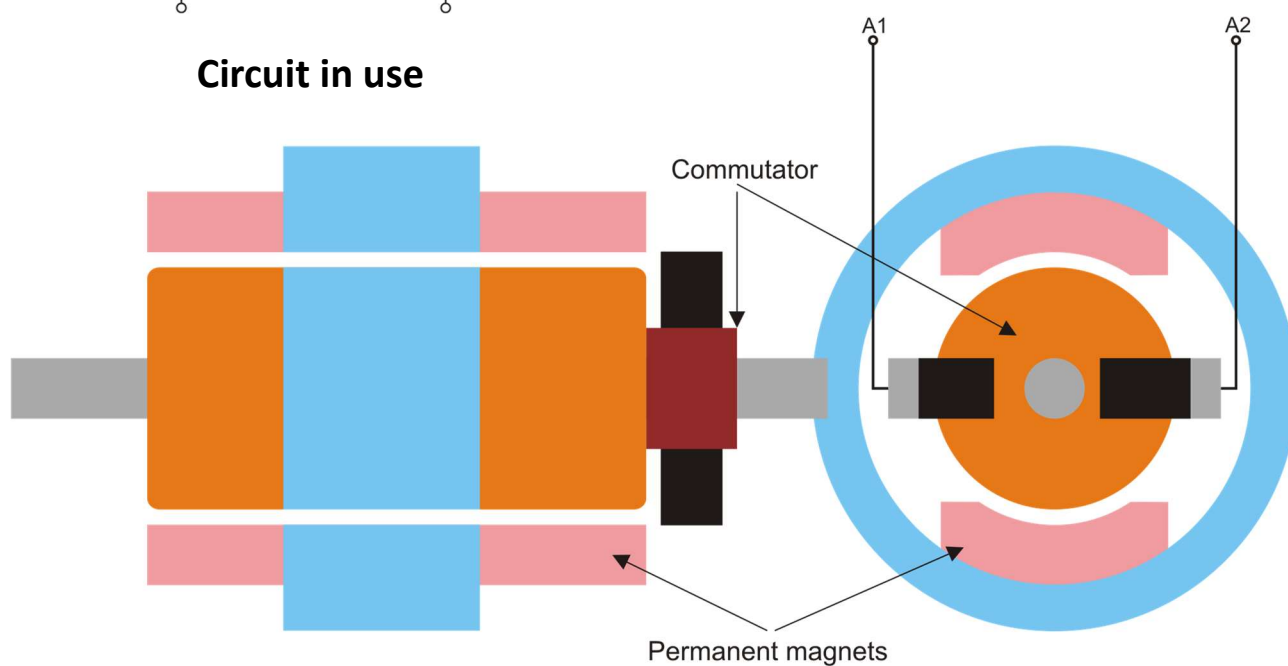
V+

V-

Colour

Red

Black

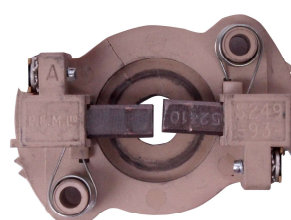
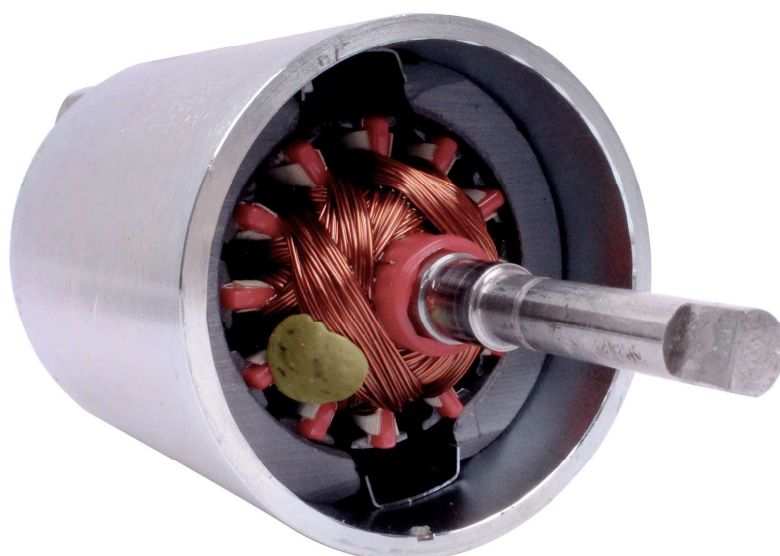


Construction

Reference

The dynamometer

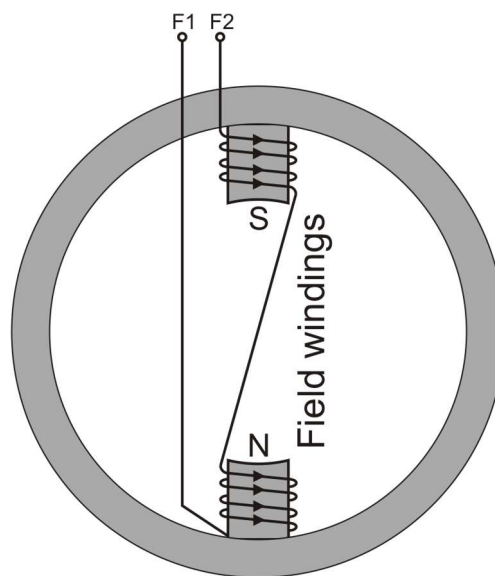
Modern electrical machines



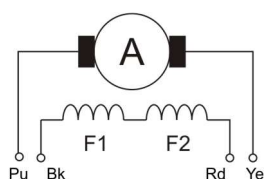
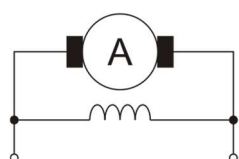
Reference

The shunt motor

Modern electrical machines

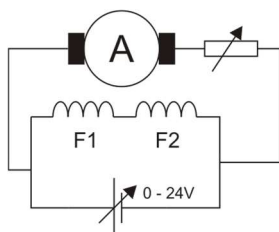
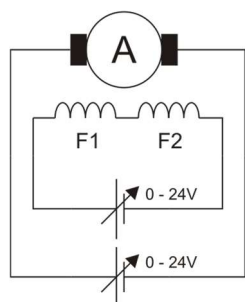


Stator winding circuit



Conventional symbol

Actual circuit



Circuit in use

Connection

Armature

Armature

Field coil

Field coil

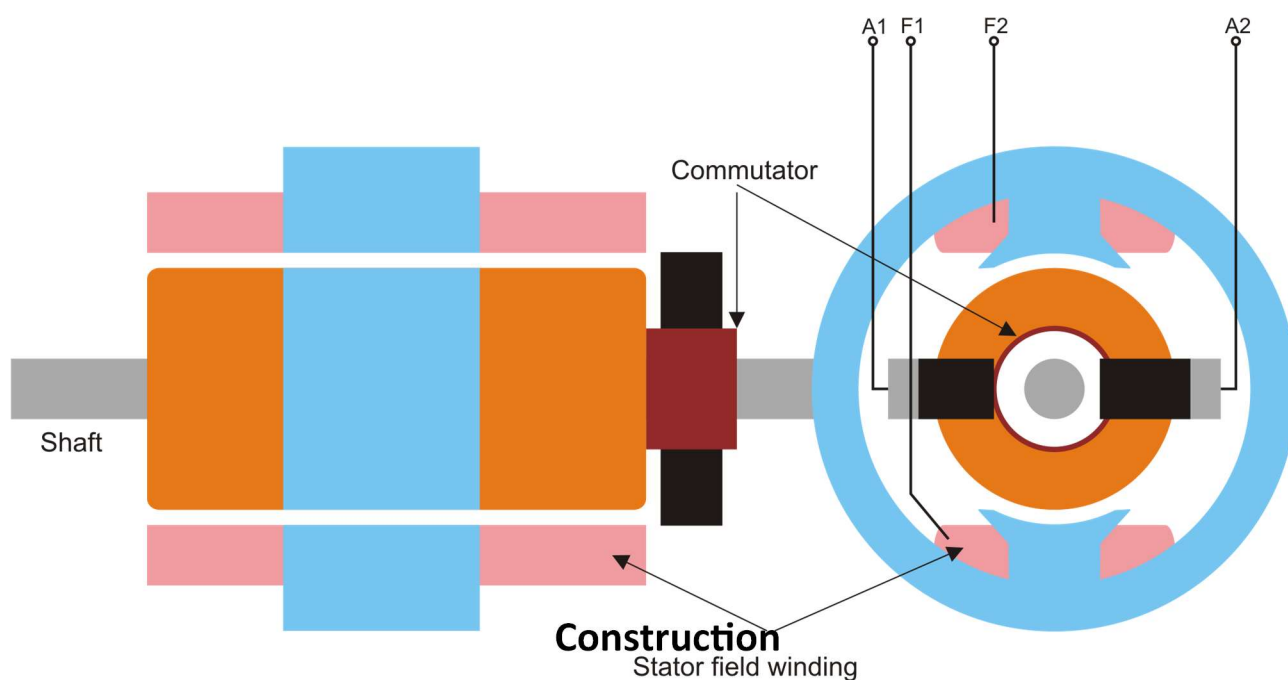
Colour

Purple

Yellow

Black

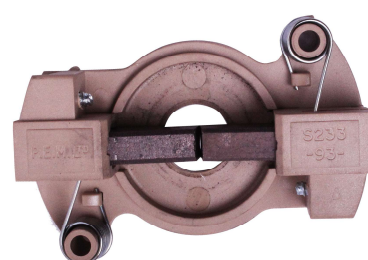
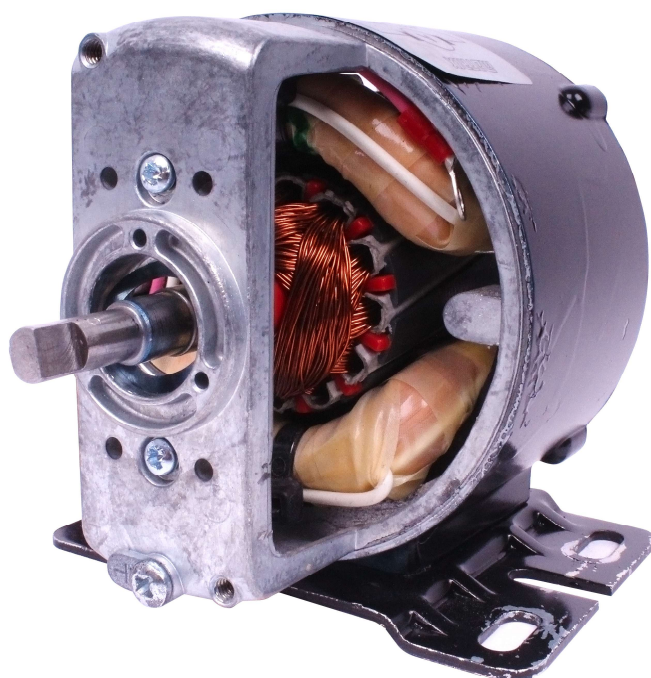
Red



Reference

The shunt motor

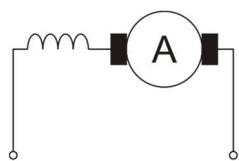
Modern electrical machines



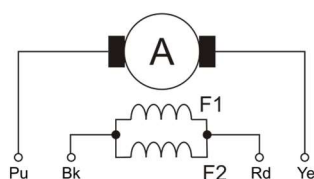
Reference

The series motor

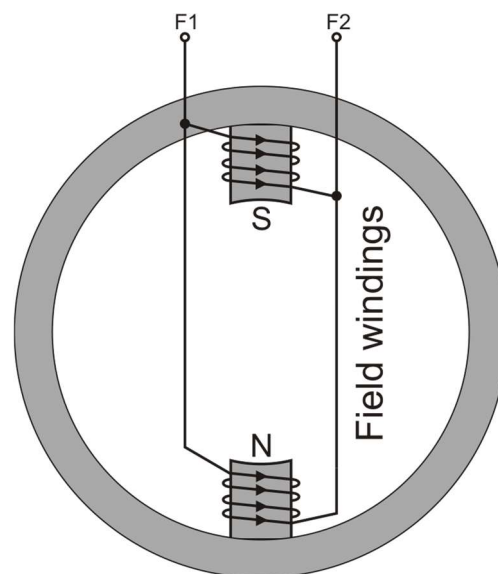
Modern electrical machines



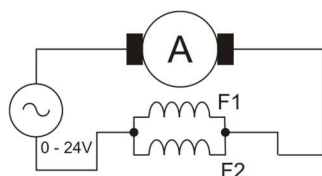
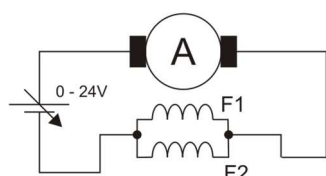
Conventional symbol



Actual circuit



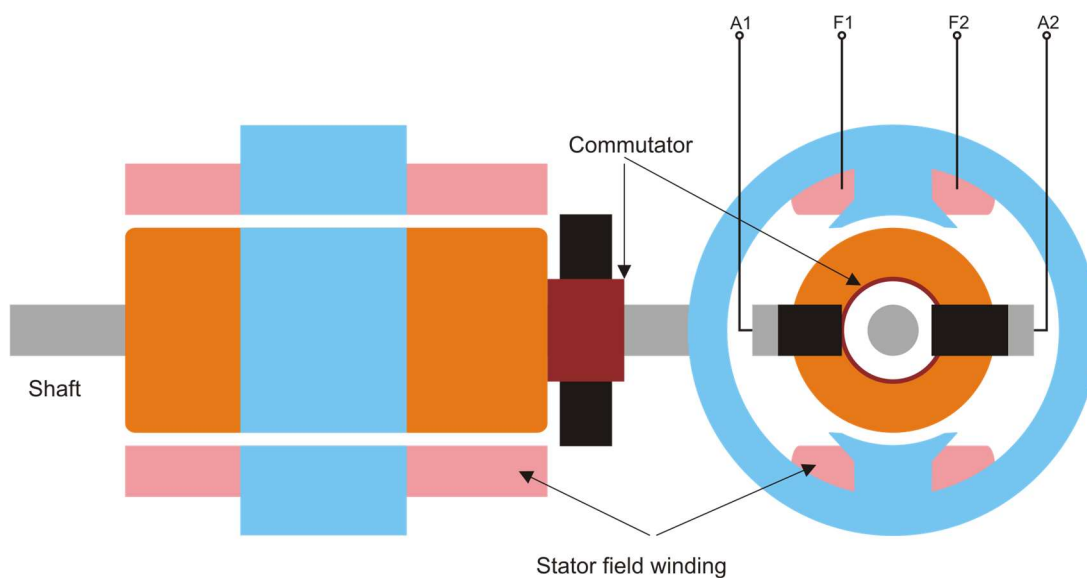
Stator winding circuit



Circuit in use

Connection
Armature
Armature
Field coil
Field coil

Colour
Purple
Yellow
Black
Red

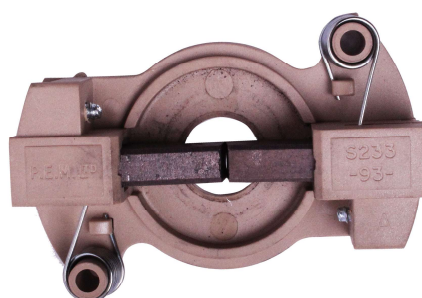
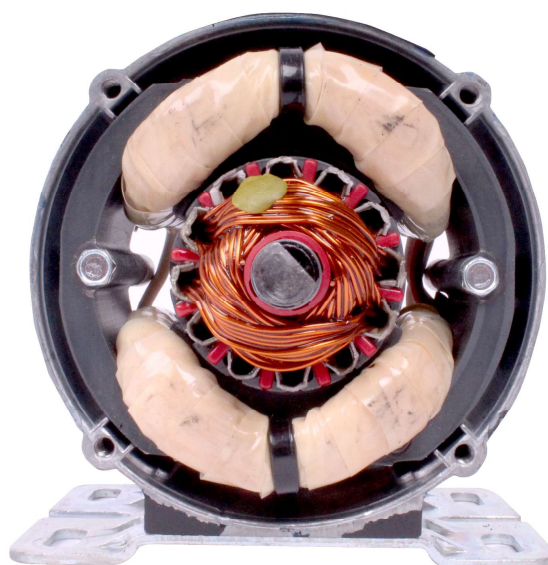


Construction

Reference

The series motor

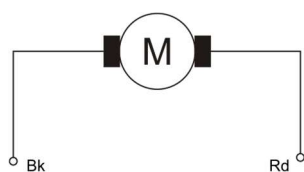
Modern electrical machines



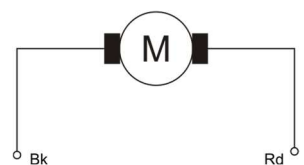
Reference

The DC motor

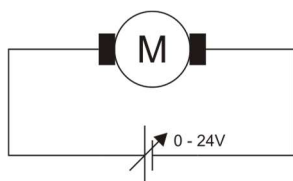
Modern electrical machines



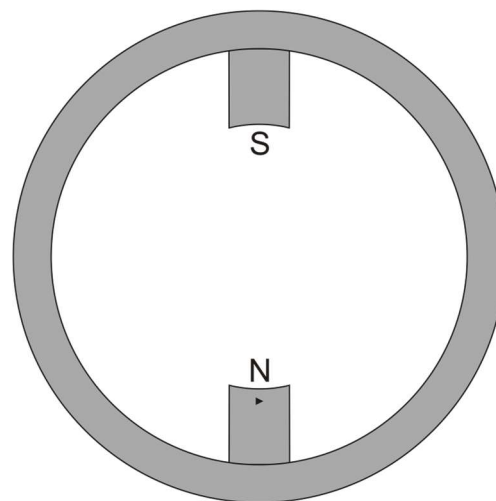
Conventional symbol



Actual circuit



Circuit in use



Stator winding circuit

Connection

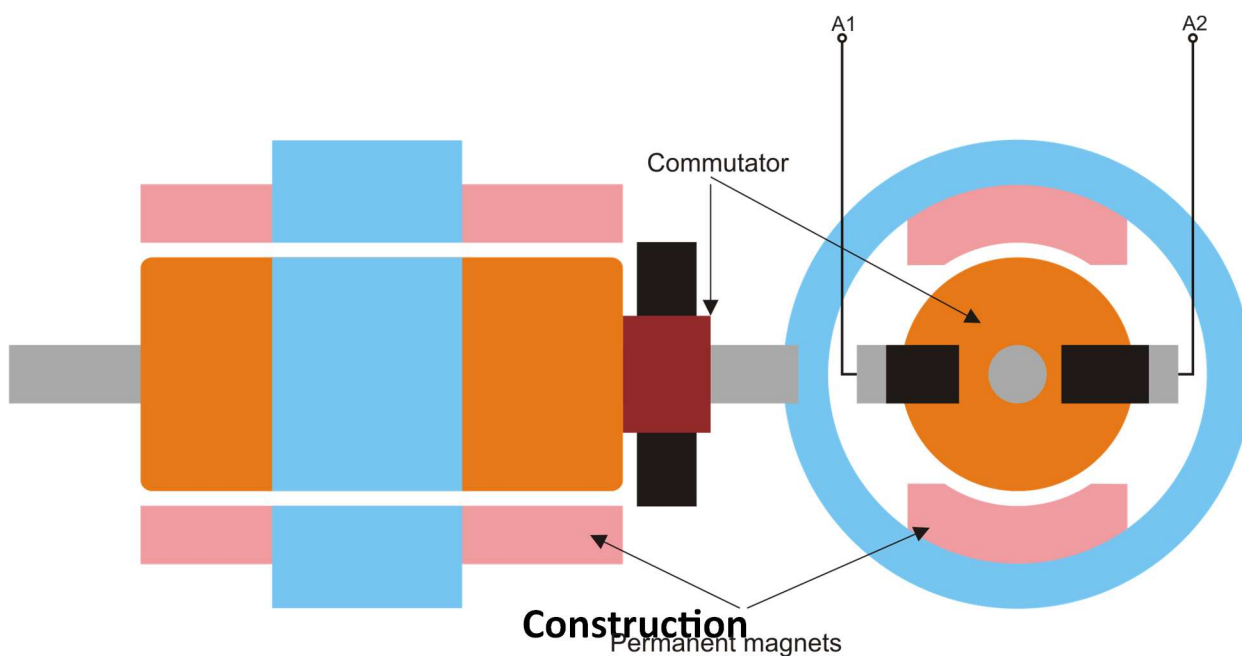
V+

V-

Colour

Red

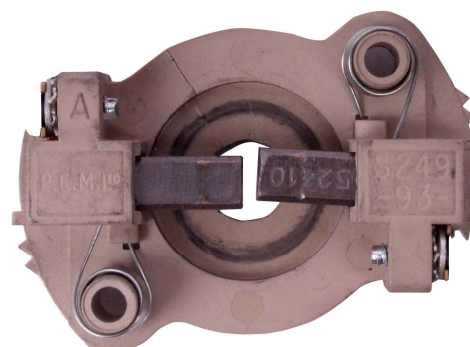
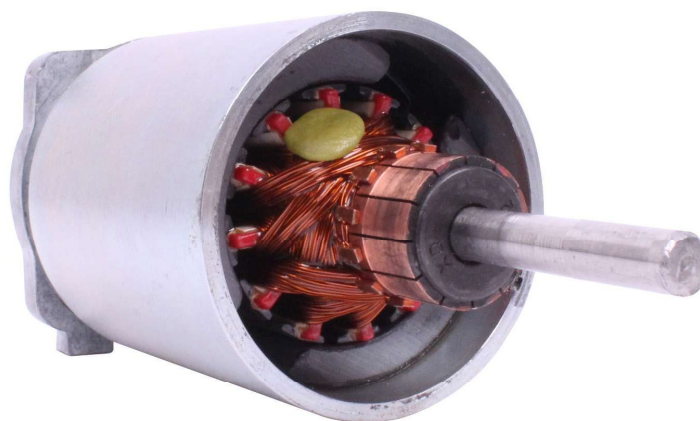
Black



Reference

The DC motor

Modern electrical machines



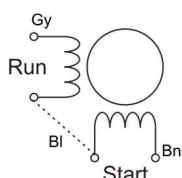
Reference

The single phase AC induction motor

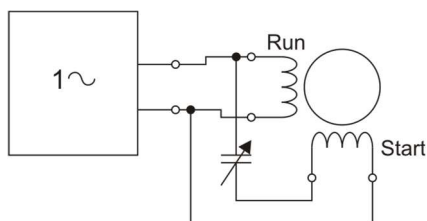
Modern electrical machines



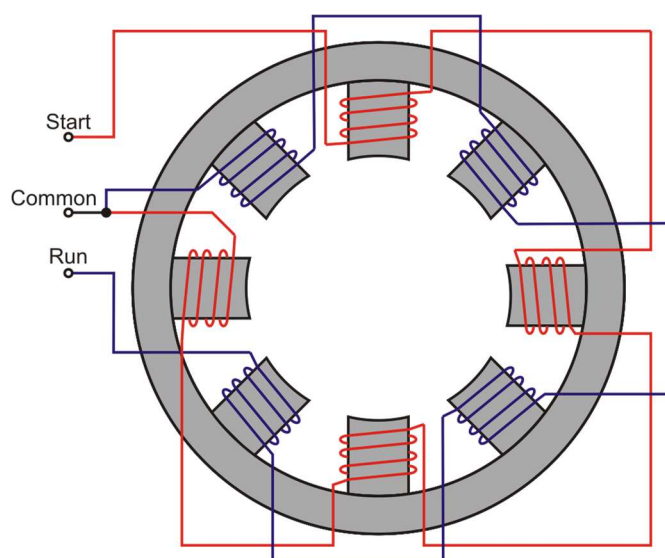
Conventional symbol



Actual circuit

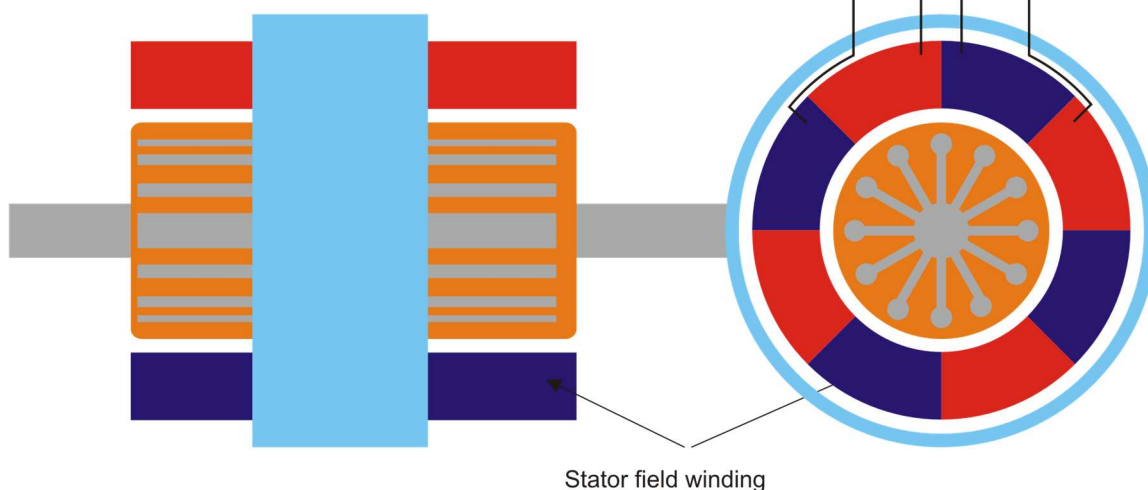


Circuit in use



Stator winding circuit

Connection	Colour
Run	Grey
Start	Brown
Common	Blue



Stator field winding

Construction

Reference

The single phase AC induction motor

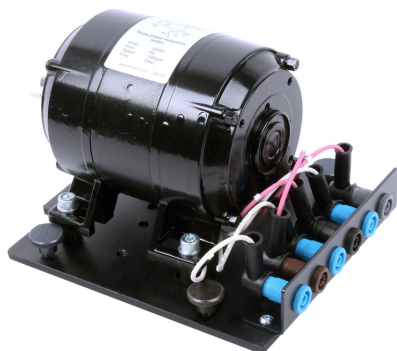
Modern electrical machines



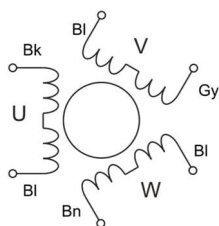
Reference

The three phase AC induction motor

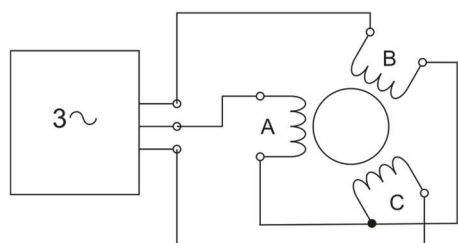
Modern electrical machines



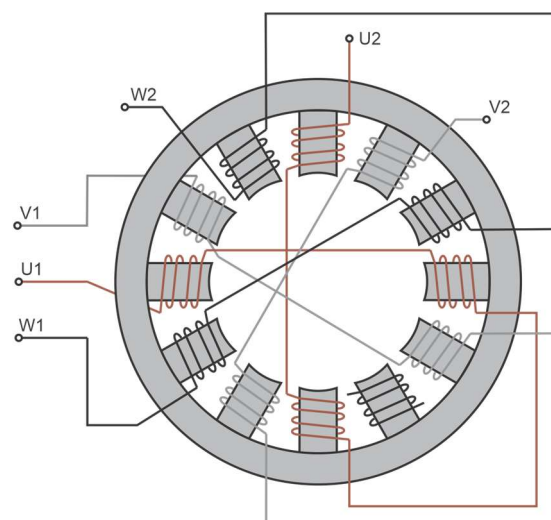
Conventional symbol



Actual circuit



Circuit

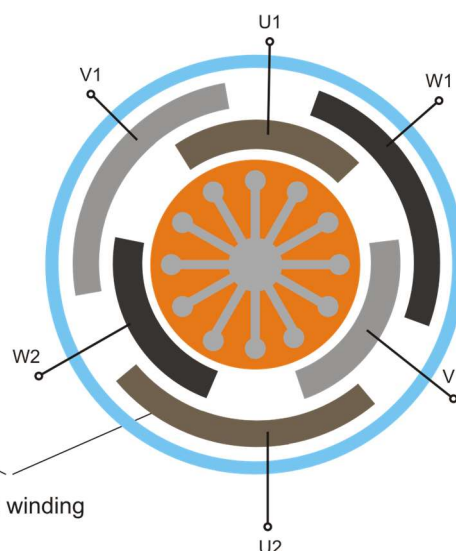


Stator winding circuit

Connection	Colour
Phase U1	Black
Phase U2	Blue
Phase V1	Grey
Phase V2	Blue
Phase W1	Brown
Phase W2	Blue



Stator field winding

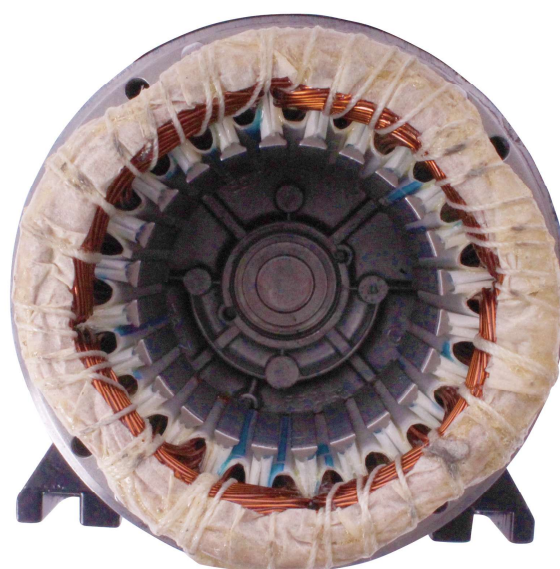
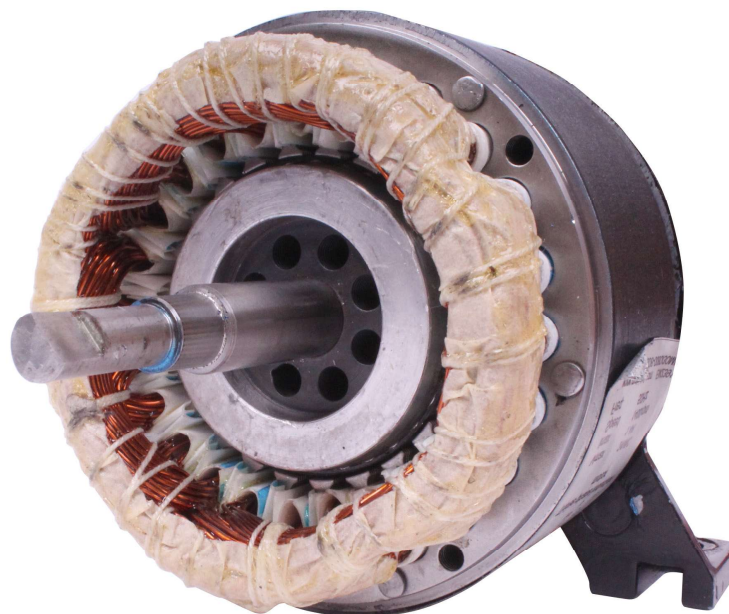


Construction

Reference

The three phase AC induction motor

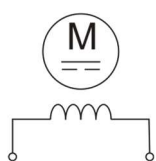
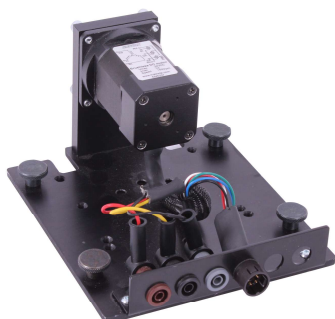
Modern electrical machines



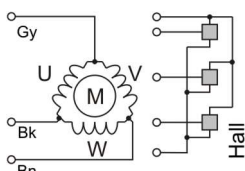
Reference

The Brushless DC motor

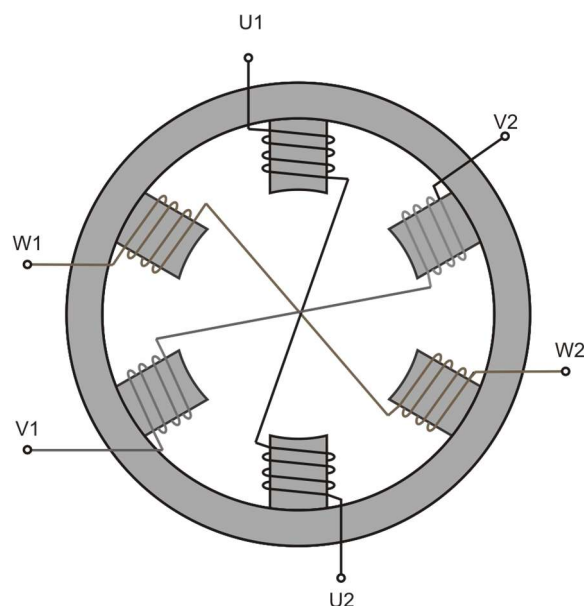
Modern electrical machines



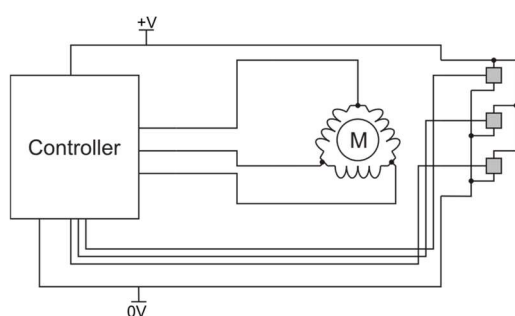
**Conventional
symbol**



Actual circuit



Stator winding circuit



Circuit in use

Connection

Phase U

Phase V

Phase W

Colour

Black

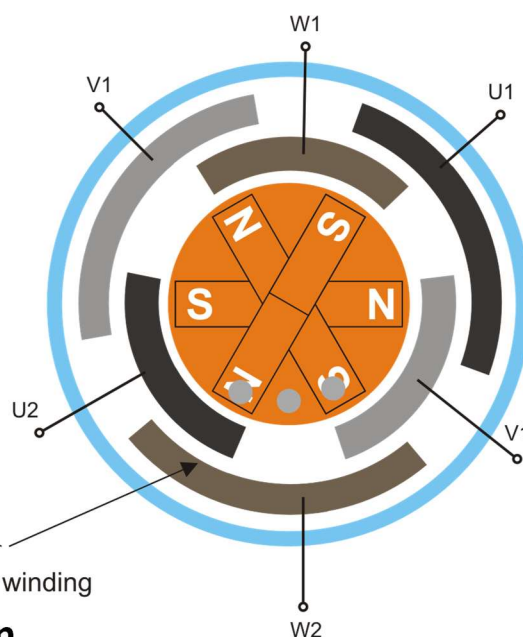
Grey

Brown



Stator field winding

Construction



Reference

The Brushless DC motor

Modern electrical machines

